Implementation of FastSpeech 2-Based Text-to-Speech Synthesis

Technical Project Report

This project implements separate FastSpeech 2 text-to-speech models for English and Telugu.

1. Project Objective

The objective is to develop neural text-to-speech systems for English and Telugu using the FastSpeech 2 architecture. The systems generate speech from input text and analyze the predicted duration, pitch and energy representations.

2. Datasets and Data Splitting

Language	Dataset	Training samples	Validation samples	Training hours	Validation hours
English	Svarah	3380	691	4.7052	0.9682
Telugu	IndicTTS Telugu	2613	525	5.0021	1.0023

The English split contains approximately 4.71 training hours and 0.97 validation hours. The Telugu split contains approximately 5.00 training hours and 1.00 validation hours. The split metadata is retained with the project.

3. Model Architecture

FastSpeech 2 is a non-autoregressive text-to-speech architecture. It uses a text encoder to represent input phonemes, a variance adaptor to incorporate duration, pitch and energy information, and a mel-spectrogram decoder to generate acoustic features. A postnet refines the generated mel-spectrogram. A vocoder converts the mel-spectrogram into a speech waveform.

Processing flow: Input text → Text preprocessing and phoneme sequence → FastSpeech 2 encoder → Variance adaptor → Mel decoder and postnet → Vocoder → Speech audio.

4. Training Configuration and Checkpoints

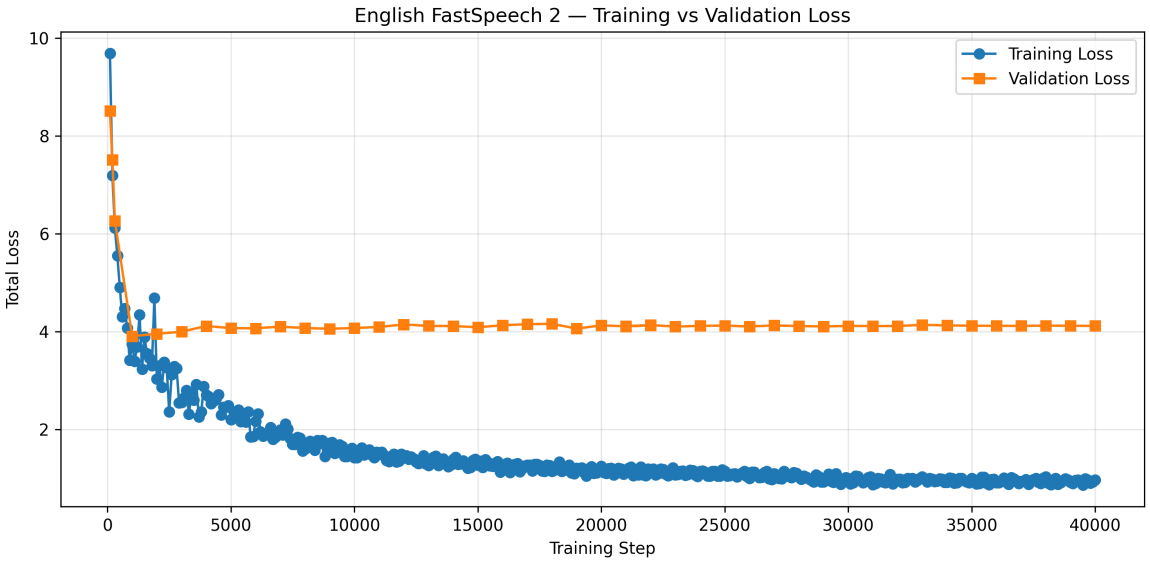
Separate configurations and checkpoints are maintained for English and Telugu. The final available checkpoints are at training step 40,000. The trained models were loaded for inference; no additional training was performed during final testing.

The project includes the language-specific configuration files, training and validation logs, and final model checkpoints.

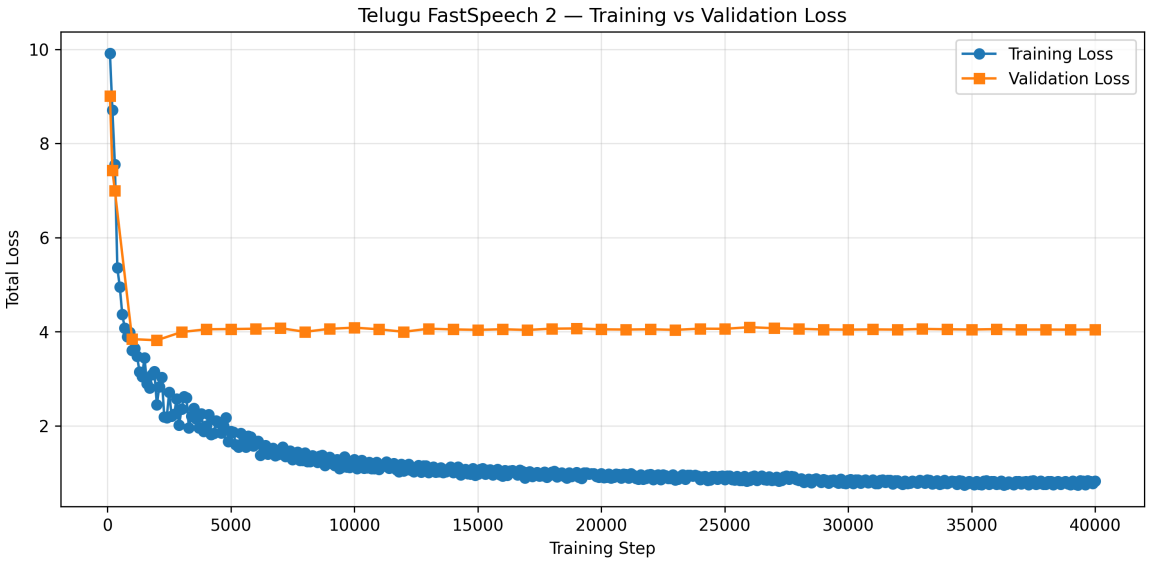
5. Training and Validation Results

Training and validation total-loss curves were generated from the recorded log entries. These plots show the logged checkpoints rather than every individual training step.

English Loss Curve



Telugu Loss Curve



6. Speech Synthesis Evaluation

Five test sentences were used for each language. The generated outputs are stored as WAV audio files. The English test set includes three selected test sentences and two sentences taken from the Svarah validation transcripts. The Telugu test set contains five Telugu sentences.

The generated English samples were reported as generally clear, although some pronunciation variation was noticed. The Telugu samples were reported as clear during listening checks.

7. Variance Adaptor Analysis

Predicted duration, pitch and energy values were extracted for each test sentence. Separate plots were generated for the English and Telugu models. The corresponding prediction data is also saved in JSON format.

The pitch and energy plots represent model-output units. They should not be interpreted as calibrated physical measurements in Hertz or energy units because the original inference normalization statistics were unavailable.

8. Project Deliverables

The submission includes the English and Telugu model checkpoints, configuration files, source code, dataset split metadata, training and validation logs, loss curves, five generated test audios per language, variance analysis JSON files, and duration, pitch and energy plots.

9. Conclusion

The project implements separate FastSpeech 2-based speech synthesis pipelines for English and Telugu. Both trained checkpoints were used to generate five test audio samples per language. Predicted duration, pitch and energy were analyzed, and the associated plots and model artifacts were retained for submission.